

Economic Narrative Indices and Media-Based Sentiment Measures: A Systematic Review of Methodologies, Applications, and Research Gaps (2007–2025)

Abstract

This systematic review synthesizes evidence from 50 peer-reviewed studies (2007–2025) examining how sentiment extracted from textual data informs economic forecasting. We document clear methodological evolution from dictionary-based approaches to Large Language Models, establishing that sentiment-based models improve forecast accuracy by 10–30% over benchmarks for GDP, inflation, stock volatility, and recession prediction. However, critical gaps persist: 84% of studies focus on English-language data from the US and Europe; no economic sentiment indices exist for Bangla or most South-Asian languages; and causal mechanisms remain underspecified. This review identifies three research frontiers: (1) geographic expansion to underrepresented regions and languages, (2) theoretical formalization of narrative mechanisms, and (3) methodological standardization. We propose the Bangla Economic Narrative Index (BENI) as a case study for bridging these gaps.

Keywords: Sentiment analysis, economic narratives, natural language processing, forecasting, machine learning

JEL Codes: C53, C55, E37, E44, G12, G17

1 Introduction

Traditional economic forecasting relies on quantitative statistics published with long lags (30–90 days) and low frequency. Meanwhile, real-time textual data continuously reveal economic sentiment: media narratives about economic conditions, corporate disclosures, and public expectations. This gap has motivated substantial research: from Tetlock (2007)’s analysis of Wall Street Journal sentiment predicting stock returns, to recent LLM-based decomposition of macroeconomic narratives (Kwon et al., 2025).

Despite 18 years of research and methodological sophistication, critical gaps remain. *Geographic bias is stark:* 56% of studies focus on the United States, 26% on Europe, less than 5% on emerging markets. *Linguistic coverage is narrow:* 84% analyze English-language text; zero studies construct

economic sentiment indices for Bangla (265M speakers), Hindi (600M), or most non-OECD languages. *Theory is weak*: while narrative economics increasingly emphasizes sentiment transmission, empirical sentiment research remains descriptive rather than mechanistic.

This systematic review addresses four questions: (1) What methodologies have been employed? (2) Where does sentiment add forecasting value? (3) How robust are current measures? (4) What are critical research gaps? We synthesize 50 empirical studies to establish: (i) comprehensive methodological taxonomy; (ii) quantitative synthesis of forecast improvements; (iii) explicit documentation of geographic/linguistic gaps; (iv) proposal for expanding sentiment analysis to underrepresented regions (specifically, Bangla-speaking markets).

2 Methodology

2.1 Search and Inclusion

Following PRISMA guidelines, we searched Web of Science, Scopus, EconLit, SSRN, and Federal Reserve working papers using: (“sentiment analysis” OR “text analysis” OR “narrative index”) AND (“forecasting” OR “economic”) AND (“machine learning” OR “NLP”).

Inclusion: Studies (2007–2025) that (1) extract sentiment/narratives from text, (2) apply to economic/financial forecasting, (3) employ systematic validation. Exclusion: pure methodology papers, non-applied work, forecasting without textual analysis.

Initial 387 records → 275 after deduplication → 98 screened → 50 final. Quality assessed on methodological rigor, data quality, validation robustness, reproducibility.

3 Methodological Evolution

3.1 Dictionary-Based Methods (2007–2016)

Tetlock (2007) pioneered lexicon-based approaches using the Harvard IV-4 dictionary on Wall Street Journal columns, finding media pessimism predicts downward stock pressure. Recognition that general dictionaries misclassify financial language (“liability” marked negative) motivated domain-specific alternatives. Loughran and McDonald (2011) developed a finance-specific dictionary improving accuracy 15–25%.

Trade-off: Interpretability vs. context sensitivity. Dictionary methods remain valuable for reproducibility and require no training data.

3.2 Machine Learning Era (2016–2023)

Random Forests, SVM, and neural networks gain traction. Hong et al. (2023) find Random Forests superior for inflation forecasting, capturing non-linear relationships. Deep learning approaches (LSTM, CNN) added but require substantial training data (5,000–50,000 labeled examples typical).

3.3 Transformer Models (2020–2024)

BERT and variants dominate. Mishev et al. (2020) report Matthews Correlation Coefficient of 0.895 using BART, outperforming simpler methods. FinBERT, fine-tuned on financial corpus, and InflaBERT (Talazadeh & Peraković, 2024) achieve state-of-the-art but require computational resources.

3.4 Large Language Models (2023–Present)

LLMs (GPT-4, GPT-5) enable zero-shot sentiment decomposition. Kwon et al. (2025) decomposed 200,000 Wall Street Journal articles into demand vs. supply drivers with 92% accuracy. Advantages: no training data, superior context understanding. Limitations: hallucination risk, cost, opacity.

Key Insight: Progression shows interpretability-performance trade-off: dictionaries most transparent, LLMs most accurate.

4 Empirical Performance

4.1 Improvements by Domain

Target	Horizon	Improvement	Key Studies
GDP / Industrial Production	Q / Monthly	10–50% RMSE	Thorsrud 2016, Barbaglia 2024
Inflation (CPI)	6–12 months	20–30% RMSE	Hong et al. 2023, Eugster & Uhl 2024
Stock Volatility	1–6 months	Strong (VIX)	Da et al. 2015, Audrino 2024
Equity Returns	1–6 months	30% RMSE	Tetlock 2007, Ren et al. 2018
Unemployment	Monthly	Weak (<10%)	Ho et al. 2021
Consumer Confidence	Monthly	20%	Aguilar et al. 2021
Recession Probability	6–12 months	80%+ accuracy	Huang et al. 2018

Table 1: *Sentiment Forecast Performance.* Based on 50 reviewed studies. Medium-term horizons (6–12 months) most successful.

Strong results concentrate in volatility, recession prediction, and investment. Weak results for unemployment suggest labor market less sentiment-responsive. *Timeliness advantage:* sentiment indices publish daily vs. quarterly official data. Crisis periods amplify value: Aguilar et al. (2021) show Spanish news sentiment captured COVID-19 recession immediately.

5 Geographic and Linguistic Coverage

5.1 Regional Distribution

Region	Studies	%	Critical Gaps
North America	28	56%	—
Europe	13	26%	Southern/Eastern Europe
East Asia	4	8%	Indonesia, Thailand, Vietnam
South Asia	2	4%	India, Bangladesh, Pakistan
Latin America	0	0%	Brazil, Mexico
Africa	0	0%	All economies
Middle East	0	0%	All economies

Table 2: *Geographic Concentration*. Severe underrepresentation of emerging markets and non-English-speaking regions.

Critical gaps: Africa (zero studies, 1.4B population); Latin America (zero, G-20 member Brazil unrepresented); South Asia (minimal despite 2B population). **English-language bias:** 84% of studies analyze English text; 12% use machine translation; 4% native-language analysis.

5.2 The Bangla Case

With 265M native speakers (Bangladesh, India, diaspora), Bangla ranks 7th globally. Yet *zero studies construct economic sentiment indices for Bangla*. Recent resources exist: BanglaBERT (Google), SentiBangla lexicon, MuRIL multilingual model—but remain unapplied to economic forecasting. Bangladesh has dense newspaper ecosystem (Prothom Alo, The Daily Star, Dhaka Tribune), growing financial markets, and increasingly digital economy. Missed opportunity for regional sentiment analysis and novel research.

6 Validation and Robustness

6.1 Standard Approaches

42/50 studies (84%) employ out-of-sample validation. Rolling window estimation standard (28 studies). Diebold-Mariano tests used in 35 studies. Most compare multiple benchmarks: naive models, time-series, traditional indicators.

6.2 Limitations

Short samples: Many studies cover 1–2 business cycles post-2000. Few historical studies (Cal-dara & Iacoviello 2022 exception: 1900–2019). Implication: results may be cycle-specific, not generalizable.

Causality unresolved: Does sentiment *cause* outcomes or merely reveal them earlier? Tetlock (2007) found return reversals suggesting noise; others interpret as causal. Few use instrumental variables or natural experiments. Reverse causality difficult to rule out.

7 Critical Gaps

7.1 Causal Mechanisms

Core unresolved question: sentiment transmission mechanisms largely black-box. How do narratives aggregate from individual to macro level? What determines which narratives "go viral"? How do policy communication and media narratives interact? Largely unexplored.

7.2 Theoretical Integration

Most studies empirically driven without formal theory. Exceptions (Angeletos & La'O 2013, Shiller 2017) rarely integrated. Need: agent-based models with narrative contagion, DSGE models with sentiment shocks, information diffusion networks.

7.3 Methodological Standardization

Heterogeneity in preprocessing, aggregation, validation hampers meta-analysis. No consensus on optimal topic numbers, dictionary selection, LLM prompting. Publication bias likely (positive results more publishable).

8 Proposed Research Agenda

8.1 1. Geographic and Linguistic Expansion

Priority: Develop sentiment indices for India, Pakistan, Bangladesh, Vietnam, Nigeria, Brazil.

Case Study: Bangla Economic Narrative Index (BENI)

- Data: 8–10 major Bangla newspapers (2018–2024)
- Methods: BanglaBERT + topic modeling (keyATM) + dynamic factor model
- Targets: Bangladesh GDP (quarterly), inflation (monthly), exchange rate
- Novelty: First native-language economic sentiment index for South Asia
- Validation: Lead-lag analysis, Granger causality, MIDAS regression

Requires: digitized news corpora, domain-specific lexicons, native transformer fine-tuning, validation against local economic data.

8.2 2. Theoretical Formalization

- Agent-based models with narrative transmission
- DSGE models with sentiment shocks and heterogeneous expectations
- Information diffusion networks linking media → public → economic behavior
- Micro-foundations: how do individuals form/update economic narratives?

8.3 3. Methodological Standardization

- Benchmark dataset for comparing methods
- Agreed validation protocol and performance metrics
- Meta-analysis of dictionary vs. ML vs. LLM across tasks
- Pre-registration to reduce publication bias

9 Conclusion

Sentiment analysis has matured from niche curiosity to credible forecasting tool. Evidence firmly establishes predictive power for volatility, recession probability, inflation expectations. Yet stark geographic and linguistic bias, weak theoretical grounding, and methodological fragmentation persist.

This review identifies two immediate priorities: (1) *geographic expansion*—extending proven methodologies to Africa, Latin America, South Asia, particularly leveraging recently developed multilingual NLP tools; (2) *formal theory*—embedding narrative dynamics into canonical economic models.

The proposed Bangla Economic Narrative Index exemplifies this agenda: applying established techniques to new linguistic context, validating on new economic data, potentially discovering context-dependent narrative-economy relationships.

Future research should prioritize closing geographic gaps through international collaboration, open-source tool development, and theoretical innovation. The next decade will reveal whether narrative-based economic analysis becomes standard in forecasting and policy institutions globally, or remains accessible only to well-resourced developed-country institutions.

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